

Personal data

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Academic appointments

Current position

September 15, 2026 – Current **postdoctoral researcher**, *Max Planck Institute of Molecular Cell Biology and Genetics (MPI-CBG)*, Dresden, Germany, Group of Algebraic Systems Biology
Group Leader: Heather Harrington.

Education

November 1, 2022 – June 11, 2026 **PhD in Engineering and Information Science in Information Engineering and Science**, *University of Siena*, Siena, Italy, Thesis: “*Product of Tensors and the Algebra of Networks*”, Advisor: Prof. Luca Chiantini,
Final evaluation: Excellent cum laude.

October 1, 2019 – April 22, 2022 **Master Degree in Applied Mathematics**, *University of Siena*, Italy, 110/110 cum laude.
Thesis: “*Combinatorial and Discrete Differential Operators for the Study of Beta-sheets*”, Advisor: Prof. Cristiano Bocci.

October 1, 2015 – April 12, 2019 **Bachelor Degree in Mathematics**, *University of Siena*, Italy, 110/110 cum laude.
Thesis: “*Operatori Logici e Reti Neurali*”, Advisor: Prof. Luca Chiantini.

September 2010 – July 2015 **Upper secondary school diploma**, *Liceo Scientifico “Galileo Galilei” of Siena*, Italy, 100/100 cum laude.

Awards, Grants and Fellowships

March 16, 2026 **Winner for the local 3-Minute-Thesis Competition 2026 of PhD students at the University of Siena**, *Academic competition from Coimbra Group in which PhD students present their research results to a general audience in less than three minutes*, University of Siena, Siena, Italy.

November 1, 2022 – October 31, 2025 **Ph.D. Full Scholarship in Engineering and Information Science in Engineering and Science (XXXVIII cycle)**, *University of Siena*, Siena, Italy.

July 7 – 11, 2025 **Travel award for the 2025 SIAM Conference on Applied Algebraic Geometry (AG25) (invited as a mini-symposium speaker)**, *University of Wisconsin-Madison*, Madison, Wisconsin, USA.

November 23, 2022 **Best student in Applied Mathematics at DIISM in a.y. 2020-2021 Awarding**, *University of Siena*, Siena, Italy.

Research interests

My research lies at the interface of algebraic geometry, tensor theory, and network models, with applications ranging from algebraic statistics to machine learning and computational biology.

- **Bhattacharya–Mesner Tensor Product and Algebra of Networks.** Development of an algebraic framework to describe signal propagation in networks through tensor representations and the Bhattacharya-Mesner product, leading to a tensorial description of network dynamics (see [1]).
- **Tensor Rank and Polynomial Decompositions.** Study of tensor rank and related decomposition problems, including Waring-type decompositions of polynomials and recent work on binomial rank (see [8]). This project was developed during my visiting period at the Université Toulouse III – Paul Sabatier.
- **Algebraic Statistics and Phylogenetic Networks.** Application of tensorial and algebraic methods to the study of Markov models and phylogenetic networks, with emphasis on distinguishing network structures via algebraic invariants. This project was developed during my visiting period at the Auburn University, Alabama.
- **Geometric and Topological Methods in Machine Learning.** Use of tools from topological data analysis to investigate the geometry and topology of machine learning models, including the study of training dynamics of generative adversarial networks (see [2]).
- **Computational Biology.** Development of geometric and morphological methods for the analysis of protein structures, including algorithms for computing atom depth and molecular descriptors (see [5], [7]).

Publications

Journal articles

- [1] L. Chiantini, G. A. D’Inverno, S. Marzali, , *Product of tensors and description of networks*, *Advances in Applied Mathematics* **180**, 103117 (2026),
doi: 10.1016/j.aam.2026.103117.
- [2] B. T. Corradini, B. Cullen, C. Gallegati, G. A. D’Inverno, S. Marzali, M. Bianchini, F. Scarselli, *Training dynamics of GANs through the lens of persistent homology*, *Neurocomputing* **661**, 131976 (2026),
doi: 10.1016/j.neucom.2025.131976.
- [3] S. Marzali, *Boolean operators and neural networks*, *Annali dell’Università di Ferrara* (2024),
doi: 10.1007/s11565-024-00541-5.

- [4] M. Tanfoni, E. G. Ceroni, S. Marziali, N. Pancino, M. Maggini, M. Bianchini, *Generated or Not Generated (GNG): The Importance of Background in the Detection of Fake Images*, *Electronics* **13**, no. 16, 3161 (2024), doi: 10.3390/electronics13163161.
- [5] S. Marziali, G. Nunziati, A. L. Prete, N. Niccolai, M. Bianchini, *SADIC v2: A modern implementation of the Simple Atom Depth Index Calculator*, *SoftwareX* **27** (2024), doi: 10.1016/j.softx.2024.101803.
- [6] F. Guerranti, M. Mannino, F. Baccini, P. Bongini, N. Pancino, A. Visibelli, S. Marziali, *CaregiverMatcher: graph neural networks for connecting caregivers of rare disease patients*, *Procedia Computer Science* **192** (2021), 1696 – 1704, doi: 10.1016/j.procs.2021.08.174.

Conference Proceedings

- [7] S. Marziali, G. Nunziati, A. L. Prete, N. Niccolai, S. Brunetti, M. Bianchini, *A Discrete Geometry Method for Atom Depth Computation in Complex Molecular Systems*, *International Conference on Discrete Geometry and Mathematical Morphology* (2024), 443 – 455,

Submitted

- [8] L. Chiantini, F. Gesmundo, S. Marziali, *Waring decompositions of special binomials*, arXiv preprint arXiv:2512.13805 (2025), doi.org/10.48550/arXiv.2512.13805.

Teaching experience

Bachelor and master level

- June 3 – July 23, 2026 **Teacher of Discrete Mathematics and Theory 2**, *CET Academic Programs for the University of Virginia (UVA)*, Siena, Italy.
- May 12 – 23, 2025 **Teaching for the “Ciclo seminariale di matematica in lingua inglese” - Centro CLUSS - UniStraSi**, *University for Foreigners of Siena (UniStraSi)*, Siena, Italy, 12 hours.
- October 1, 2022 – July 31, 2023 **Teaching Assistant for the Mathematical Analysis 2 undergraduate course (Spring 2023)**, *Department of Information Engineering and Mathematics, University of Siena*, Siena, Italy.
- October 1, 2022 – June 30, 2023 **Teaching Assistant for the Linear Algebra undergraduate course (Fall 2022, Spring 2023)**, *Department of Information Engineering and Mathematics, University of Siena*, Siena, Italy.
- October 1, 2022 – June 30, 2023 **Teaching Assistant for the Mathematical Analysis 1 undergraduate course (Fall 2022)**, *Department of Information Engineering and Mathematics, University of Siena*, Siena, Italy.

Secondary school level

- December 21, 2023 – April 6, 2024 **Teacher of Mathematics, Logic and Physics for the “Corso di preparazione ai test di ammissione e chirurgia a.a. 2023/2024”**, *Department of Medicine, Surgery and Neuroscience, University of Siena*, Siena, Italy.

Research Stays

- October 17, 2024 – April 30, 2025 **Visiting scholar**, *Université Toulouse III - Paul Sabatier*, Toulouse, France
Advisor: Prof. Fulvio Gesmundo.
- January 12 – April 4, 2024 **Visiting scholar**, *Auburn University*, Auburn, Alabama, USA
Advisor: Prof. Luke Oeding.

Conferences, workshops and seminars

Conference and workshop presentations

- September 2, 2026 **INABAG 2026 - School and Conference**, *Lecturers of the school: Kathlén Kohn (KTH, Sweden), Frédéric Holweck (U. Tech. de Belfort Montbéliard, France)*, Contribution title: "Waring decompositions of special binomials", University of Siena, Siena, Italy
August 31 – September 4, 2026.
- July 14, 2026 **TENORS Workshop 2 – Computational Methods for Tensors in Mathematics and Quantum Physics**, Contribution title: "Waring decompositions of special binomials", University of Trento, Trento, Italy
July 13 – 17, 2026.
- July 10, 2026 **Effective Methods in Algebraic Geometry (MEGA) 2026**, Contribution title: "Neural Learning of Fast Matrix Multiplication Algorithms: A StrassenNet Approach", Durham University, Durham, United Kingdom
July 6 – 10, 2026.
- October 3, 2025 **WORKSHOP ARCIDOSSO25: Aspetti matematici e fondazionali della modellizzazione di sistemi complessi**, Contribution title: "Product of Tensors and Description of Networks", Castello Aldobrandesco, Arcidosso (Grosseto), Italy
October 3 – 5, 2025.
- 8 July, 2024 **SIAM Conference on Applied Algebraic Geometry (AG25), mini-symposium "Tensors in Applications"**, Contribution title: "The Batthacharya-Mesner Product: From Tensor Rank Analysis to Description of Networks", University of Wisconsin-Madison, Madison, USA
July 7-11, 2025.
- May 14, 2024 **JM Invariant at 60 - A Conference on Tensor Invariants in Geometry and Complexity Theory**, Contribution title: "Product of Tensors and Description of Networks", Auburn University, Auburn, Alabama, USA
May 13-17, 2024.
- April 18, 2024 **IAPR Third International Conference on Discrete Geometry and Mathematical Morphology**, Contribution title: "A Discrete Geometry Method for Atom Depth Computation in Complex Molecular Systems", University of Florence, Florence, Italy
April 15-18, 2024.

Conference and workshop attendance

- July 29 – August 2, 2024 **Effective Methods in Algebraic Geometry – MEGA 2024**, MPI für Mathematik in den Naturwissenschaften Leipzig and Universität Leipzig, Leipzig, Germany.
- June 13, 14, 2024 **II Meeting UMI for Doctoral Students**, Naples, Italy.
- September 18 – 22, 2023 **Third Young Applied Mathematicians Conference**, *University of Siena*, Siena, Italy.
- January 24, 25, 2023 **Workshop on Communicating Mathematics for the Public**, Isaac Newton Institute, Cambridge, UK.

Seminars

- October 23, 2025 **“Product of tensors and description of the network for matrix multiplication via Strassen’s algorithm”**, within *“TensorDec group Seminars”*, University of Trento, Trento, Italy.
- May 29, 2024 **“Product of Tensors and Description of Networks”**, within *‘TensorDec group Seminars’*, University of Trento, Trento, Italy.
- March 19, 20, 21, 2024 **“Product of Tensors and Description of Networks”**, *Johns Hopkins University*, Baltimore, Maryland, USA.
- January 30, 2024 **“Product of Tensors and Description of Networks”**, within *“Auburn Algebra Seminar”*, University of Auburn, Auburn, Alabama, USA.
- November 17, 2023 **“Product of Tensors and Description of Networks”**, within *“Applied Algebraic Geometry 2023-2024 Seminars”*, University of Florence, Florence, Italy.

Organiser

- September 22 – 26, 2025 **Organising and Scientific Committee for “The Fifth Young Applied Mathematicians Conference”**, *University of Padua*, Padua, Italy, with G. Auricchio, E. Bachini, C. Carrara, G. A. D’Inverno, C. Faccio, A. Kushova, A. Marchetti, C. Millevoi, E. Onofri.
- December 4 – 6, 2024 **Organising Committee for the “Learning on Graph Conference 2024 – Italy Meetup”**, *University of Siena*, Siena, Italy, with P. Bongini, F. Costanti, B. Cullen, G. A. D’Inverno, C. Gallegati, C. Graziani, V. Lachi, N. Pancino, F. Pichi.
- September 16 – 20, 2024 **Organising and Scientific Committee for the “Fourth Young Applied Mathematicians Conference”**, *University of Rome “La Sapienza”*, Rome, Italy, with G. Auricchio, C. Carrara, G. A. D’Inverno, C. Graziani, A. Kushova, G. Loli, A. Marchetti, M. Menci, E. Onofri.

Educational activities

- October 13 – 24, 2025 **Masterclass and PhD course - Tensors and their applications: foundations and recent advances**, *University of Trento*, Lecturer: Jeroen Zuiddam (U. Amsterdam / QuSoft), Exercise sessions: Maxim van den Berg (Ruhr-U. Bochum / U. Amsterdam), Trento, Italy.

- May 26 – 30, 2025 **TiME 2025 – Invariant Theory from Classic to Modern Developments**, *Lie-Størmer Center*, Tromsø, Norway.
- June 10 – 14, 2023 **6th Advanced Course on Data Science & Machine Learning (ACDL)**, *Castiglione della Pescaia (Grosseto)*, Italy.
- July 22 – August 16, 2019 **Interuniversity Mathematics Summer School (SMI)**, *Course of Numerical Analysis with Professor Michele Benzi (Scuola Normale Superiore, Pisa, Italy): final grade A; Course of Differential Geometry with Prof. Liviu Ornea (University of Bucharest, Romania): final grade B*, Perugia, Italy.

Further information

Scientific Associations and Membership

- *Society for Industrial and Applied Mathematics (SIAM)*, Student Membership (July 1, 2025 – December 31, 2026).

Professional activities and projects

- May 9, 2022 – May 20, 2026 **Pint of Science**, *Local Team Collaborator*, Siena, Italy.
- October 5, 19, 2023 **7th Rare Disease Hackathon 2020: Health System Forum - Re-evolution of Intelligence Special Lysosomal Storage Diseases**, *Challenge: "Caregiving pathways. Patients, caregivers and lysosomal storage disorders: challenges and solutions"*, Proposed solution: "SAM: Assistance and Monitoring System", "Stazione La Leopolda", Florence, Italy
Third place awarded.
- November 4, 5, 2020 **Rare Disease Hackathon 2020 (digital edition)**, "Stazione La Leopolda", Firenze, Italy
Finalist Team.

Computer skills and competencies

- Matlab, Python, Latex
- European Computer Driving Licence (ECDL Certificate) achieved on June 3, 2014.

Language competences

- Italian: Mother tongue
- English: Proficient, Cambridge C1 Advanced exam in English (CAE), Certification from Cambridge University, English Qualifications, June 2024.